a

ROC curves, IxIDN-Matched (n = 4,364; 50% positive)

True positive rate

False positive rate

AUROC (all 4,364 pairs)

- DisAtlas (full) 0.765
- DisAtlas (unseen in learning) 0.742
- Claude-Chat Opus 5.5 0.692
- Claude-Sci Opus 5.5 0.685
- Phylo Biomni 0.627
- TxGNN (published ckpt) 0.566
- Gemini Pro 3.1 0.497

b

Ranking metrics (dot = value, bar = 95% CI, pair bootstrap)

	AUPRC	F1 at k = #pos.	Precision, top 10%	NPV, bottom 10%
DisAtlas (full)	0.777	0.708	0.917	0.780
DisAtlas (unseen in learning)	0.757	0.680	0.913	0.782
Claude-Chat Opus 5.5	0.739	0.628	0.933	0.635
Claude-Sci Opus 5.5	0.708	0.640	0.860	0.638
Phylo Biomni	0.664	0.596	0.830	0.500
TxGNN (published ckpt)	0.606	0.541	0.773	0.530
Gemini Pro 3.1	0.536	0.497	0.631	0.489

random

c

Run-to-run agreement of pair scores

one point = one pair of runs; TxGNN (single run) not shown

Spearman ρ between two runs

DisAtlas (full) | DisAtlas (unseen in learning) | Claude-Chat Opus 5.5 | Claude-Sci Opus 5.5 | Phylo Biomni | Gemini Pro 3.1

d

Overlap of top-10% pairs (Jaccard; 436 of 4,364 per method)

	DisAtlas (full)	DisAtlas (unseen in learning)	Claude-Chat Opus 5.5	Claude-Sci Opus 5.5	Phylo Biomni	TxGNN (published ckpt)	Gemini Pro 3.1
DisAtlas (full)	1.00	0.63	0.26	0.18	0.20	0.11	0.13
DisAtlas (unseen in learning)	0.63	1.00	0.22	0.14	0.14	0.12	0.10
Claude-Chat Opus 5.5	0.26	0.22	1.00	0.51	0.26	0.12	0.14
Claude-Sci Opus 5.5	0.18	0.14	0.51	1.00	0.26	0.09	0.15
Phylo Biomni	0.20	0.14	0.26	0.26	1.00	0.06	0.19
TxGNN (published ckpt)	0.11	0.12	0.12	0.09	0.06	1.00	0.02
Gemini Pro 3.1	0.13	0.10	0.14	0.15	0.19	0.02	1.00

Jaccard index